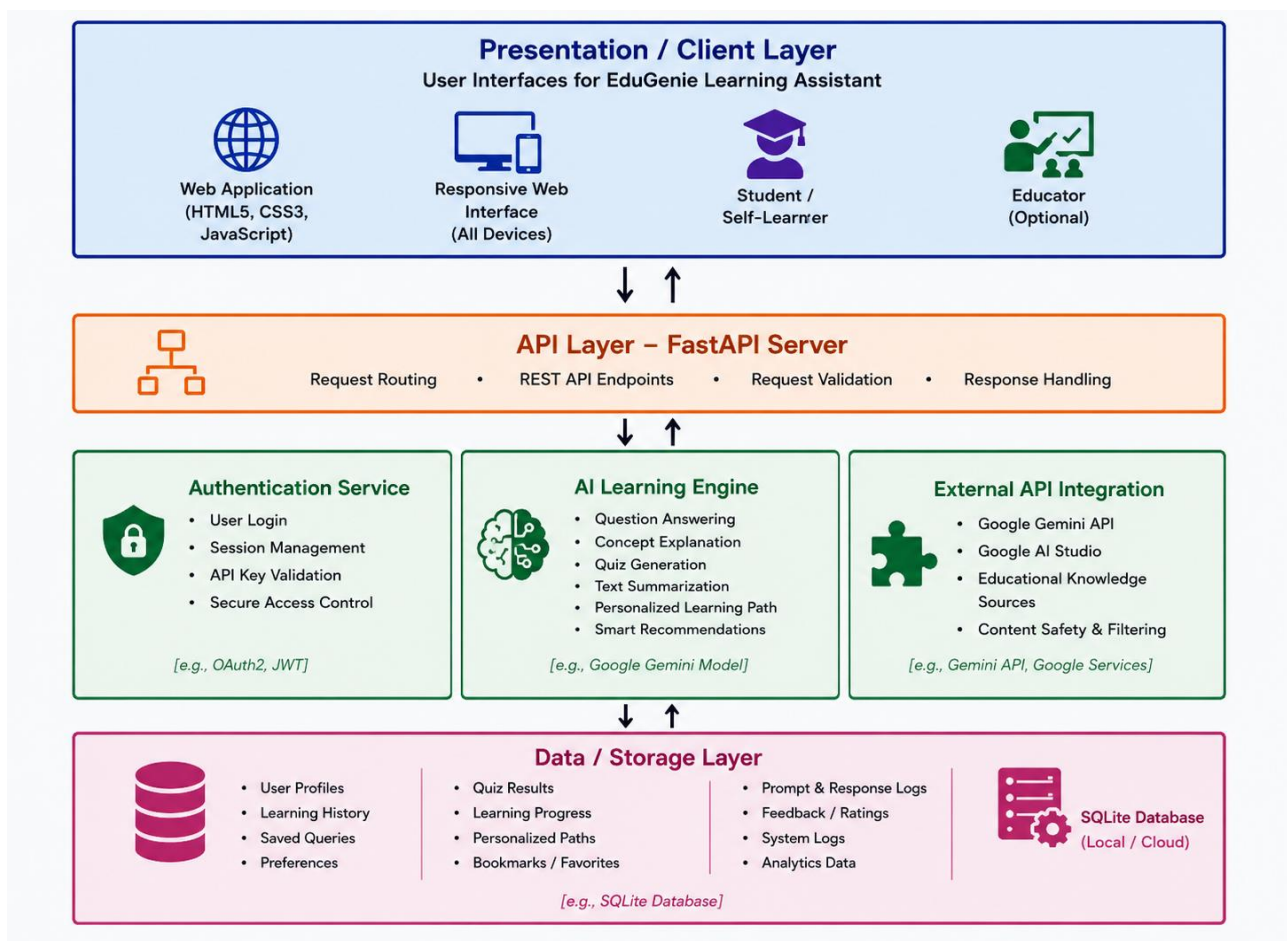


Solution Architecture

Date	6 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive and responsive user interface where students and educators can ask questions, generate quizzes, summarize educational content, and view personalized learning paths. It collects user input and displays AI-generated educational responses.	HTML5, CSS3, JavaScript
API Gateway / Application Layer	Receives user requests from the frontend, validates the input, routes requests to the appropriate services, communicates with the Google Gemini API, and returns AI-generated responses to the client. It also manages REST API endpoints and request processing.	Python, FastAPI, Uvicorn, REST API
Core Services (AI Learning Engine / Microservices)	Implements the core business logic of EduGenie, including intelligent question answering, concept explanation, AI-powered quiz generation, educational text summarization, and personalized learning path recommendations by integrating with Google Gemini.	Google Gemini API, Google AI Studio, Python, FastAPI
Database / Data Layer	Stores user profiles (optional), learning history, quiz results, saved queries, learning progress, prompt-response logs, and system information to support personalized learning and future analysis.	SQLite, Python, SQL